

Counter-propagating edge channels in a pocket of the confining potential and the Hall response

a numerical study

Working note

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Abstract

We study numerically (Kwant, tight-binding model) a six-terminal Hall bar whose confining potential forms a well (a “pocket”) in front of the edge. Inside the pocket the next Landau level crosses the Fermi energy twice, and a pair of edge channels appears, one of which runs against the chirality. Everything that happens to such a pair between two neighbouring contacts reduces to a single number, the backward transmission probability t_b , which gives the Hall and longitudinal resistances explicitly. Quantization is broken when the counter-propagating channel reaches a contact ($t_b \neq 0$). Backscattering inside the pair restores it. With the image-charge potential of Ref. [1] the picture is the same. The two-terminal conductance on which the conclusions of [1] rest equals $\nu + t_b$. It is therefore an integer both for $t_b = 0$ and for $t_b = 1$, and it misses exactly the regime in which the Hall bar deviates most. We end with signatures of the mechanism that can be tested in a four-terminal geometry.

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1 The question

In experiments with a GaAs two-dimensional electron gas placed inside a metallic split-ring resonator, the plateaus of the quantum Hall effect were found to be distorted [3]. This was first attributed to the vacuum fluctuations of the resonator field. Ref. [1] proposes an electrostatic explanation instead. The metal of the resonator next to the mesa edge carries the image of the electron, which pulls the electron towards the edge. Together with a steep confining potential this produces a well at the edge that hosts counter-propagating (non-chiral) edge channels. The authors of [1] compute only the two-terminal conductance $G(E_F)$. The comment [2] objects that (i) the self-consistent electrostatics of a GaAs mesa does not produce a pocket, and (ii) the experiment measures R_{xx} and R_{xy} in a Hall bar, not a two-terminal conductance.

We do not address whether the pocket exists (point (i)): here the pocket is put in by hand. We ask *what it does to the Hall response if it does exist*:

1. does a counter-propagating channel on one edge spoil R_{xy} and R_{xx} of a six-terminal Hall bar;
2. under which conditions (overlap of the partners, disorder, contact geometry);
3. how this compares with the conclusions of [1], which are drawn from the two-terminal conductance.

The resonator as a dynamical system is not considered.

2 Model

2.1 Hamiltonian and units

Square lattice with lattice constant $a = 1$ and hopping $t = 1$:

$$H = \sum_i (4t + V_i) c_i^\dagger c_i - t \sum_{\langle ij \rangle} e^{i\theta_{ij}} c_i^\dagger c_j, \quad (1)$$

where the Peierls phases θ_{ij} correspond to a flux of φ flux quanta h/e per plaquette. In the continuum limit $\hbar\omega_c \simeq 4\pi\varphi t$ and $l_B = a/\sqrt{2\pi\varphi}$. Unless stated otherwise the Fermi energy is fixed at $E_F = 0.4t$ and the field is swept over $\varphi = 0.008 \dots 0.06$ (plateaus $\nu = 4 \dots 1$, $l_B \approx 1.6 \dots 4.5 a$). The field calibration was checked against the Landau level spacing of a ribbon in the Landau gauge and against the plateau positions.

2.2 Geometry

A 300×80 bar with four probes of width 56 (much wider than the pocket and l_B) and six ideal ohmic contacts: semi-infinite leads in the same field and with the same edge profile (Fig. 1). Current flows $0 \rightarrow 3$; probes 1, 2, 4, 5 are floating. For our sign of the field, electrons go around the perimeter as $0 \rightarrow 5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0$.

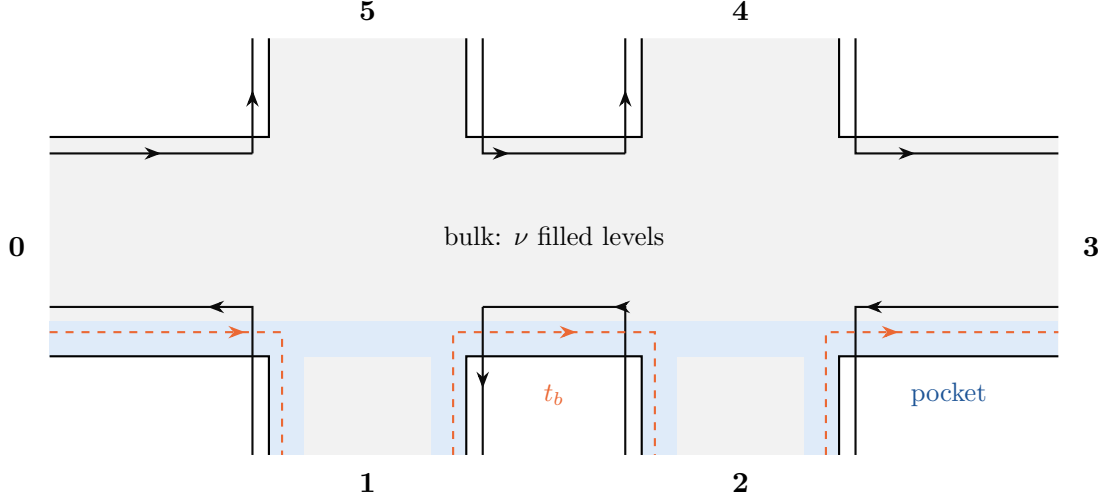


Figure 1. The Hall bar. Black arrows: the ν chiral channels on each edge (one is drawn). Blue band: the pocket at the bottom edge (including the bottom probes and the lower halves of leads 0 and 3). Orange dashes: the counter-propagating channel at the inner border of the pocket; its co-propagating partner at the outer border is not drawn. t_b is the probability that an electron emitted by the downstream contact reaches the upstream contact against the chirality.

2.3 Edge profile

The potential depends on the distance d to the sample boundary. The wall is smooth:

$$V_{\text{wall}}(d) = V_w \left(\frac{d_0 - d}{d_0} \right)^2 \theta(d_0 - d), \quad V_w = 1.5 t, \quad d_0 = 6 a.$$

We consider two kinds of pocket.

- **Rectangular pocket** (Sec. 5): $V(d) = V_{\text{wall}}(d) - V_{\text{dip}} w(d; d_{\text{in}}, d_{\text{out}})$, where w is a smoothed window (tanh fronts of width a) and $d_{\text{in}} = 6$. *Wide pocket*: $V_{\text{dip}} = 0.3 t$, width $14 a$; the pair is separated by $\approx 5.8 l_B$ (at $\nu = 1$, $\varphi = 0.028$) with overlap < 0.01 . *Narrow pocket*: $V_{\text{dip}} = 0.35 t$, width $4 a$; separation $\approx 1.6 l_B$, overlap ≈ 0.57 .
- **Image-charge potential** (Sec. 6), in the form of [1]: a metal plate parallel to the edge at distance d_{edge} ,

$$V(d) = V_{\text{wall}}(d) - E_{\text{img}} \left[\frac{1}{d + d_{\text{edge}}} - \frac{1}{d_c + d_{\text{edge}}} \right] \theta(d_c - d), \quad (2)$$

with $d_c = 40 a$ (the middle of the bar), so that the bulk and the opposite edge are untouched.

The overlap of the partners is defined as $O = \sum_y |\psi_1(y)| |\psi_2(y)|$ (1 for identical, 0 for disjoint mode wave functions of an infinite ribbon with the same profile).

Placements of the pocket: bottom edge only; both edges; and a *local* pocket on the bottom edge between probes 1 and 2 that does not touch any contact. Anderson disorder $U = 1 t$ (uniform in $[-U/2, U/2]$) fills the whole scattering region; “clean” runs have $U = 0$.

2.4 Observables

From the Kwant [5] scattering matrix we build the conductance matrix G_{ij} , drive a current I from contact 0 to contact 3 (grounded) and solve for the potentials of the floating probes. Then (all resistances in units of h/e^2)

$$R_{xy}^{(1-5)} = \frac{V_5 - V_1}{I}, \quad R_{xy}^{(2-4)} = \frac{V_4 - V_2}{I}, \quad R_{xx}^{\text{bot}} = \frac{V_1 - V_2}{I}, \quad R_{xx}^{\text{top}} = \frac{V_5 - V_4}{I}, \quad R_{2t} = \frac{V_0 - V_3}{I}.$$

For the comparison with [1] we also compute the two-terminal conductance G_{2t} of a plain 300×80 bar without probes and with the same edge profile.

3 Where the counter-propagating channel comes from

For a potential that is smooth on the scale of l_B , the Landau levels follow it, $E_n(y) \approx \hbar\omega_c(n + \frac{1}{2}) + V(y)$, and the group velocity along the edge is proportional to $\partial_y V$. Let E_F lie between levels n and $n+1$ in the bulk. If level $n+1$ dips below E_F inside the pocket, it crosses E_F twice. The outer crossing (on the wall side) gives a channel co-propagating with the chiral ones. The inner crossing, where $\partial_y V$ has the opposite sign, gives a *counter-propagating* channel. This happens on the low-field side of every plateau, as long as $(n + \frac{3}{2})\hbar\omega_c - V_{\text{dip,eff}} < E_F$ (Fig. 2). A narrow pocket gives a pair about l_B apart, a wide pocket a well-separated pair.

4 The general statement: a single parameter t_b

4.1 An edge segment

Consider a piece of the edge between neighbouring contacts a (upstream) and b (downstream). It carries ν chiral channels and the pair (co-propagating + counter-propagating). Contact a sends $\nu + 1$ channels into the segment and receives one (the counter-propagating channel); contact b sends one channel into the segment and receives $\nu + 1$. If no current leaks through the bulk to other contacts, unitarity of the segment scattering matrix gives

$$T_{a \leftarrow b} + R_b = 1, \quad T_{b \leftarrow a} + R_a = \nu + 1 \quad \implies \quad T_{b \leftarrow a} - T_{a \leftarrow b} = \nu.$$

Whatever the scattering inside the segment is (elastic backscattering, channel mixing, partial equilibration), the segment is fully described by one number, the *backward transmission* $t_b \equiv T_{a \leftarrow b} \in [0, 1]$. $t_b = 0$ is a perfect quantum Hall edge; $t_b = 1$ is a ballistic counter-propagating channel.

4.2 Formulas for the Hall bar

For a pocket on the bottom edge with the same t_b on all three bottom segments, the Landauer–Büttiker equations [4] can be solved in closed form (Table 1).

For a plain two-terminal bar whose leads carry the same edge,

$$G_{2t} = \nu + t_b^{(\text{full length})} [e^2/h], \quad (3)$$

where t_b is taken over the whole length of the edge from drain to source.

Sample edge at $\nu=1$ ($\varphi=0.028$, $l_B=2.4a$); arrows: $\downarrow$ chiral channels (towards the source), $\uparrow$ counter-propagating channel

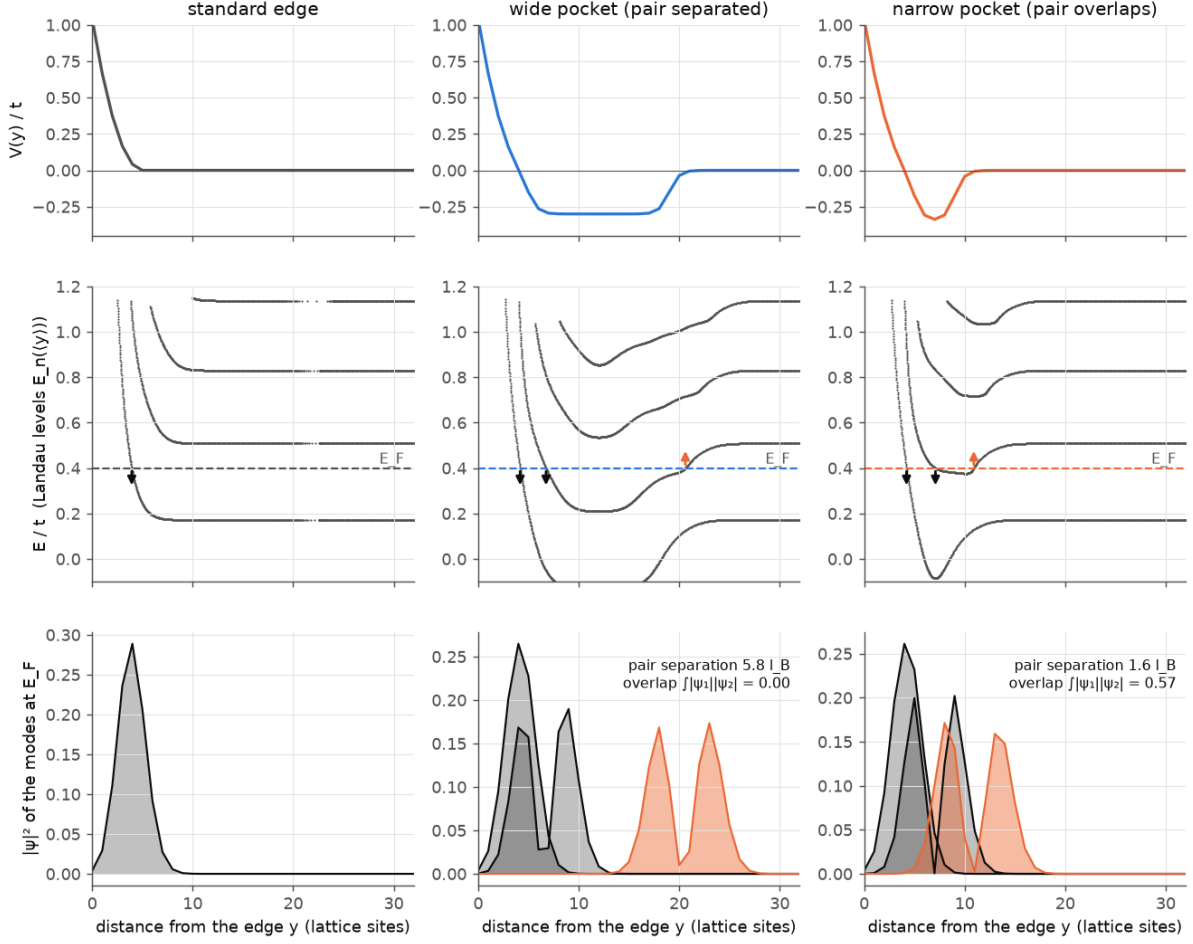


Figure 2. The edge at $\nu = 1$ ($\varphi = 0.028$). Top to bottom: the potential profile; ribbon eigenenergies against the position of the state $\langle y \rangle$ (bent Landau levels), with arrows marking the modes at E_F (black: chiral, orange: counter-propagating); profiles of the modes at E_F . Left to right: standard edge, wide and narrow pockets.

4.3 Consequences

- **Quantization is broken by the absence of backscattering, not by backscattering.** A ballistic counter-propagating channel ($t_b = 1$) carries the potential of the downstream contact back to the upstream one. This shifts R_{xy} and gives $R_{xx} \neq 0$ without any dissipation in the bulk; the bulk $\sigma_{xy} = \nu e^2/h$ stays topological.
- **Backscattering inside the pair heals.** If the partners overlap and disorder backscatters electrons, the counter-propagating channel localizes along the edge, $t_b \sim e^{-L/\xi} \rightarrow 0$, and quantization is restored.
- **A contact is needed.** If the pocket is local and does not touch any ohmic contact, the pair forms a closed loop, $t_b \equiv 0$, and there is no effect in DC transport.
- The edge without a pocket stays ideal ($R_{xx}^{\text{top}} = 0$), and the difference of the two Hall voltages equals exactly R_{xx} of the defective edge.

Table 1. Resistances of the Hall bar with a pocket on the bottom edge (in h/e^2).

quantity	formula	$\nu = 1, t_b = 1$
$R_{xy}^{(1-5)}$	$1/(\nu + t_b)$	$1/2$
$R_{xy}^{(2-4)}$	$(\nu + 2t_b)/(\nu + t_b)^2$	$3/4$
R_{xx}^{bot}	$t_b/(\nu + t_b)^2$	$1/4$
R_{xx}^{top}	0	0
R_{2t} (with floating probes)	$(\nu^2 + 3\nu t_b + 3t_b^2)/(\nu + t_b)^3$	$7/8$

5 Rectangular pocket

5.1 Pocket on one (bottom) edge

In Fig. 3 the grey curves are the standard sample: plateaus $R_{xy} = h/\nu e^2$ and $R_{xx} = 0$, with the mesoscopic fluctuations typical of a small coherent sample between the plateaus. The curves are averaged over three disorder realizations.

Pocket on the bottom edge only, disorder $U = 1 t$ (average over 3 realizations). Shading: fields with a counter-propagating pair at the bottom edge (wide pocket)

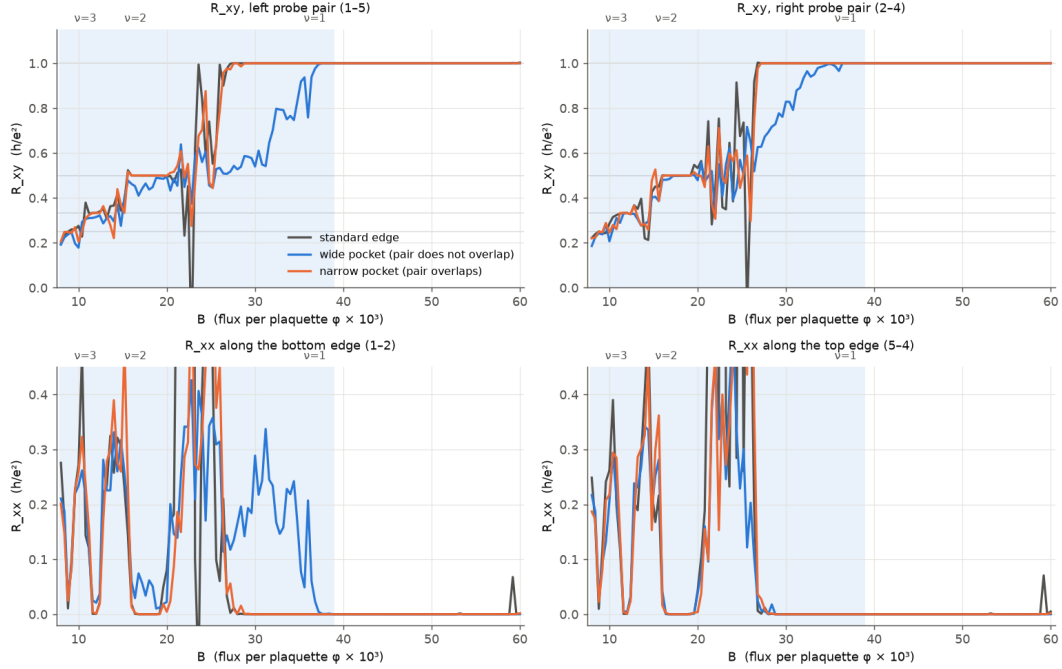


Figure 3. Field sweep at $E_F = 0.4t$, $U = 1t$, averaged over three realizations. Shading: fields at which the bottom edge carries a counter-propagating pair (wide pocket).

- **Wide pocket** (pair separated by $4-6 l_B$): on the low-field part of the $\nu = 1$ plateau ($\varphi \approx 0.027-0.038$) $R_{xy}^{(1-5)}$ drops to $0.51-0.60$ (at $\varphi \approx 0.027-0.031$; further on the deviation gradually shrinks), and $R_{xy}^{(2-4)}$ drops to ≈ 0.63 . R_{xx}^{bot} reaches ≈ 0.34 , while $R_{xx}^{\text{top}} = 0$ to within 10^{-4} at $\varphi > 0.03$. The same, weaker, is seen on the $\nu = 2$ plateau. At $\varphi > 0.039$ there is no pair and the curves coincide with the reference.
- **Narrow pocket** (separation $\approx 1.6 l_B$, overlap ≈ 0.6): the pair exists at $\varphi \approx 0.022-0.029$, partly inside the $2 \rightarrow 1$ transition. On the plateau part $\varphi \approx 0.0275-0.0285$ disorder brings R_{xy} back to h/e^2 within ≤ 0.015 , whereas without disorder $R_{xy}^{(1-5)} = 0.50-0.54$ there (Fig. 4).

5.2 Clean versus disordered

Without disorder the counter-propagating channel is ballistic, $t_b = 1$, and Kwant reproduces Table 1 *exactly*: $R_{xy}^{(1-5)} = 1/(\nu + 1)$ (0.5; 0.333; 0.25 for $\nu = 1, 2, 3$), $R_{xy}^{(2-4)} = (\nu + 2)/(\nu + 1)^2$ (0.75; 0.444; 0.3125), $R_{xx}^{\text{bot}} = 1/(\nu + 1)^2$ (0.25; 0.111; 0.0625), over the whole field range where the pair exists. This holds *for the narrow pocket too*: without scatterers the pair by itself, even an overlapping one, gives the maximal deviation. Switching on disorder in the narrow pocket restores exact quantization (Fig. 4).

Without disorder the counter-propagating channel is ballistic ($t_b = 1$) and quantization is broken; backscattering (narrow pocket + disorder) restores it

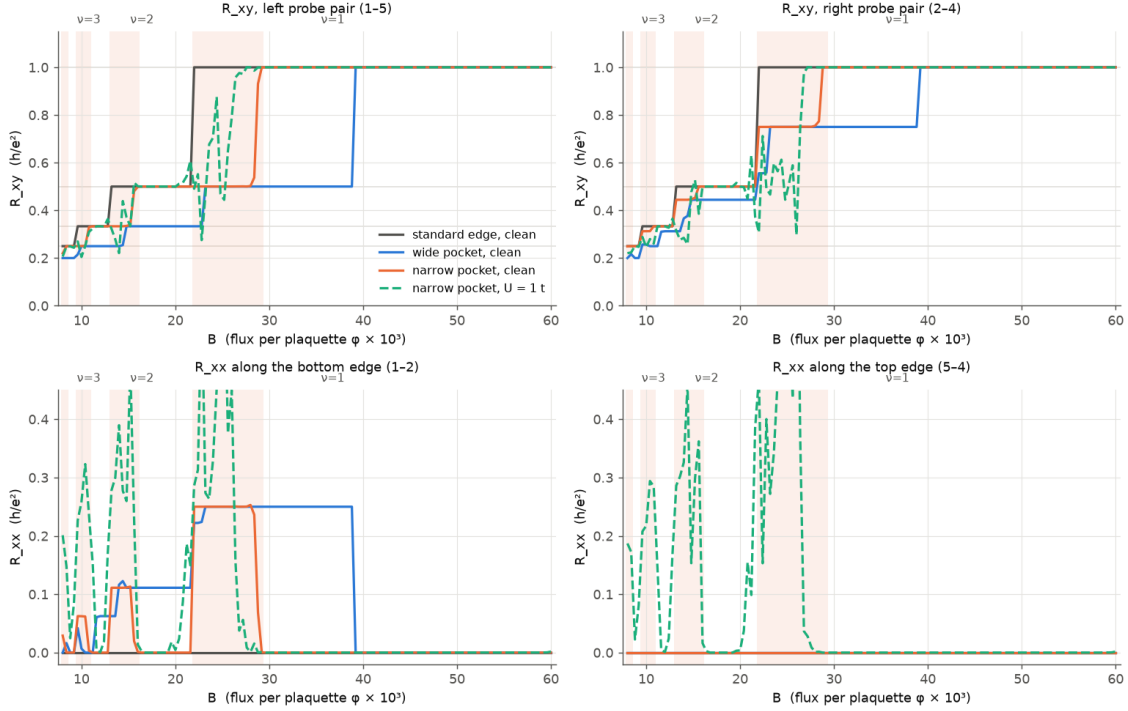


Figure 4. Without disorder ($t_b = 1$) quantization is broken according to Table 1; backscattering (narrow pocket + disorder, green dashed line) restores it.

Fig. 5 shows on the left t_b between probes 1 and 2, taken directly from the scattering matrix, and on the right the formulas of Table 1 together with Kwant points on the plateaus. In the wide pocket with disorder part of the counter-propagating flow bypasses a probe (for example $0 \rightarrow 2$ bypassing probe 1, with probability up to ≈ 0.3), so the single- t_b formula is only approximate there. All curves in the figures are computed from the full transmission matrix, not from this formula.

5.3 Do the partners overlap? The pocket width

At fixed field ($\nu = 1$, $\varphi = 0.028$), pocket depth $0.35t$ and $U = 1t$ (three realizations) we vary only the pocket width, i.e. the distance between the counter-propagating channels (Fig. 6, Table 2). While the partners overlap, disorder backscatters electrons within the pair, the counter-propagating channel localizes on a length shorter than the distance between contacts, and the response is quantized. Once the partners are separated, there is no backscattering, the counter-propagating channel reaches a contact, and quantization is broken.

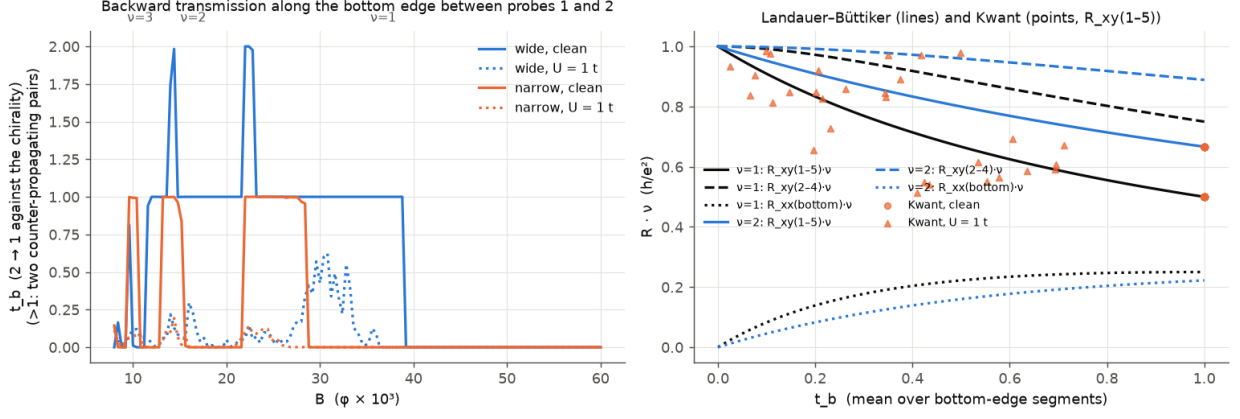


Figure 5. Left: backward transmission t_b along the bottom edge between probes 1 and 2. Right: Landauer–Büttiker formulas (lines) and Kwant (points, $R_{xy}^{(1-5)}$).

Table 2. Dependence on the pair separation ($\nu = 1$, average over three realizations).

pair separation	$\leq 2.3 l_B$	$2.7 l_B$	$3.5 l_B$	$\geq 4.3 l_B$
overlap O	0.9–0.5	0.43	0.20	≤ 0.06
t_b (segment 1–2)	≈ 0 –0.01	0.03	0.18	0.2–0.4
$R_{xy}^{(1-5)}, h/e^2$	0.97–1.00	0.94	0.75	0.53–0.58
$R_{xx}^{\text{bot}}, h/e^2$	≤ 0.03	0.06	0.19	0.15–0.27

5.4 Pockets on both edges and a local pocket

- Wide pockets on both edges: the deviations add up. For a ballistic pair the Landauer–Büttiker formulas give $R_{xy} = 1/3$, $R_{xx} = 2/9$ at $\nu = 1$; Kwant with disorder gives $R_{xy} \approx 0.36$ –0.48 on the low-field part of the plateau ($\varphi \approx 0.027$ –0.031), rising towards 1 at higher field.
- Narrow pockets on both edges with disorder: quantization survives, $|R_{xy} - h/e^2| \leq 3 \cdot 10^{-3}$ on the $\nu = 1$ plateau (one realization).
- A wide pocket only on the part of the bottom edge between probes 1 and 2, reaching no contact: the pair is closed into a loop, $t_b \equiv 0$, and on the plateaus the curves coincide with the reference, even though there is no backscattering inside the loop. On the plateaus $t_b \leq 10^{-8}$ and R_{xy} differs from the reference by at most $5 \cdot 10^{-4}$; at a single point at the plateau edge ($\varphi = 0.0276$) $R_{xx} \approx 0.08$ on both edges at once, which is leakage through the bulk. In the transitions between plateaus the curves differ mesoscopically: the pocket changes the potential landscape where the bulk states are extended.

5.5 Where the current flows

Fig. 8 shows the current density of electrons injected from probe 2. In the reference sample all of it follows the chiral path $2 \rightarrow 1$. In the clean wide pocket a second path appears: the counter-propagating channel carries part of the flow from probe 2 to the right, to drain 3, that is, against the chirality. With disorder this path is weakened but survives. In the narrow pocket with disorder only localized circulating currents around disorder resonances remain, and the counter-propagating path does not reach drain 3.

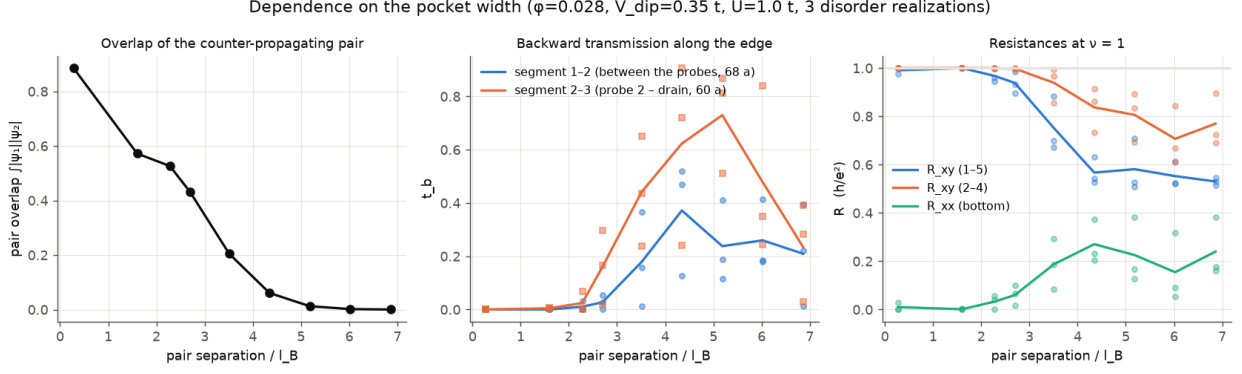


Figure 6. Dependence on the pocket width: pair overlap, backward transmission along two bottom-edge segments, and resistances at $\nu = 1$. Points: individual disorder realizations; lines: averages.

6 The image-charge potential of [1]

6.1 What Ref. [1] claims

In a single-electron model, $U_{\text{im}} = -e^2/(8\pi\epsilon|x - d_{\text{edge}}|)$ plus a steep confining potential produce a well at the edge. The energy scale is the *backscattering energy*

$$\Gamma_\ell = l_B \sqrt{\ell + 1} \partial_x U_{\text{im}}|_{\text{edge}} = \frac{e^2}{8\pi\epsilon} \frac{l_B \sqrt{\ell + 1}}{d_{\text{edge}}^2}, \quad (4)$$

the potential drop across the cyclotron orbit. When E_F is within $\sim \Gamma_\ell$ of the pocket bottom, the counter-propagating states are at most $\sim l_B \sqrt{\ell + 1}$ apart, electrons backscatter, and the two-terminal $G(E_F)$ is not quantized. The Kwant calculations of [1] (Harper–Hofstadter model, field-free leads) give only the two-terminal conductance; the Zeeman (odd) plateaus turn out to be the most fragile.

6.2 Parameters

We use the profile (2) with $d_{\text{edge}} = 18a \approx 8l_B$ at $\nu = 1$ (the same ratio d_{edge}/l_B as the estimate of [1] for the experiment) and $E_{\text{img}} = 9$. The well is then $0.22t \approx 0.58\hbar\omega_c$ deep and $\Gamma_0 = 0.036t \approx 0.1\hbar\omega_c$ at $\varphi = 0.03$: between the “realistic” estimate of [1] ($\Gamma_0/\hbar\omega_c \sim 0.02$) and the values 0.2–0.4 used in the figures of that paper. The pocket is on the bottom edge only, including the bottom probes and the leads.

In this pocket the separation of the pair is not fixed by a window width but grows with energy (Fig. 9, right). Near the pocket bottom the partners overlap ($O > 0.3$ within $\approx \Gamma_0$ of the bottom); higher up they move apart to 10–15 l_B . One pocket therefore contains both regimes of Sec. 5, at different E_F or B . In addition, the inner partner sits on the gentle $1/d$ tail, and its velocity is 4–5 times lower than in the rectangular pocket ($v \approx 0.04$ – 0.05 versus 0.16–0.21 in units of $ta/\hbar$), so disorder backscatters it even when the pair is noticeably separated.

6.3 Field sweep

At $E_F = 0.4t$ (Fig. 10):

- Without disorder we again get exactly Table 1 with $t_b = 1$ throughout the window where the pair exists: at $\nu = 1$ $R_{xy}^{(1-5)} = 1/2$, $R_{xy}^{(2-4)} = 3/4$, $R_{xx}^{\text{bot}} = 1/4$; at $\nu = 2$ and 3, $1/3$, $4/9$, $1/9$ and $1/4$, $5/16$, $1/16$. Note that $R_{xy}^{(1-5)} = 1/(\nu + 1)$ looks like a continuation of the neighbouring plateau; only $R_{xy}^{(2-4)}$ and R_{xx} of the bottom edge give the pair away.

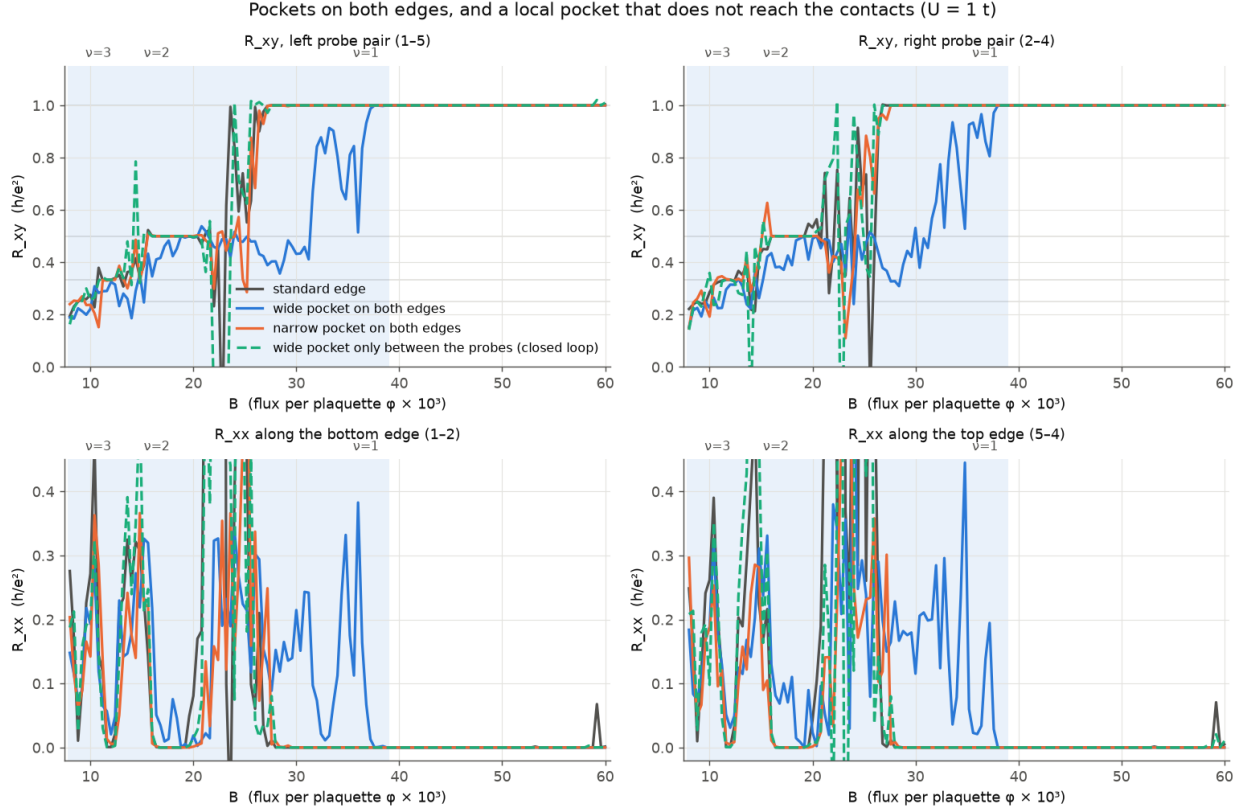


Figure 7. Pockets on both edges, and a local pocket that does not reach the contacts ($U = 1 t$).

- With disorder, on the high-field part of the window ($\varphi \approx 0.029$ – 0.032 , overlapping pair), $t_b \leq 0.11$ (mean over segments) and $R_{xy}^{(1-5)} = 0.86$ – 1.00 . On the low-field part (separated pair) $R_{xy}^{(1-5)}$ drops to 0.44 – 0.70 and R_{xx}^{bot} rises to 0.33 . This window, however, coincides with the $2 \rightarrow 1$ bulk transition of the reference, so below we repeat the comparison at fixed field.

6.4 Sweep of E_F at fixed field (the protocol of [1])

At $\varphi = 0.03$ the same sample is computed both as a Hall bar and as a plain two-terminal bar (Fig. 11, Table 3). Values with disorder are averaged over three realizations.

Table 3. E_F sweep at $\varphi = 0.03$, image-charge potential, $U = 1 t$. At $E_F < 0.19$ the bulk is empty, the top probes carry no channels, and the Hall bar cannot be measured; only G_{2t} remains.

E_F/t	pair	$G_{2t}, e^2/h$	$R_{xy}^{(1-5)}$	R_{xx}^{bot}	reference R_{xy}, R_{xx}
0.000–0.048 ($\nu = 0$)	0.8–3.2 l_B	≤ 0.003 (clean: 1)	—	—	$G_{2t} = 0$
0.052–0.12 ($\nu = 0$)	3.4–7.9 l_B	$0.07 \rightarrow 0.98$	—	—	$G_{2t} = 0$
0.372–0.412 ($\nu = 1$)	0.5–2.8 l_B , $O=0.85$ – 0.41	1.000	0.97 – 1.00^a	$\leq 0.03^a$	1, 0
0.416–0.432 ($\nu = 1$)	3.0–3.9 l_B , $O=0.34$ – 0.11	0.986 – 1.000	0.81 – 0.91	0.09 – 0.19	$1, \leq 10^{-4}$
0.436–0.448 ($\nu = 1$)	4.2–4.9 l_B , $O \leq 0.08$	0.89 – 1.035	0.73 – 0.92	0.09 – 0.29	$1, \leq 0.016$
0.372–0.49, $U = 0$		2	0.50	0.25	

^a one point: $R_{xy} = 0.88$, $R_{xx} = 0.12$.

- **The $\nu = 0$ window is the analogue of Figs. 3 and S1 of [1].** Above the pocket bottom $G_{2t} < 1$ and grows towards 1 as the pair separates. In our long sample with strong disorder this window is wider than Γ_0 , and inside it $G_{2t} = 0$ (the counter-propagating channel is fully localized) rather than partially suppressed.

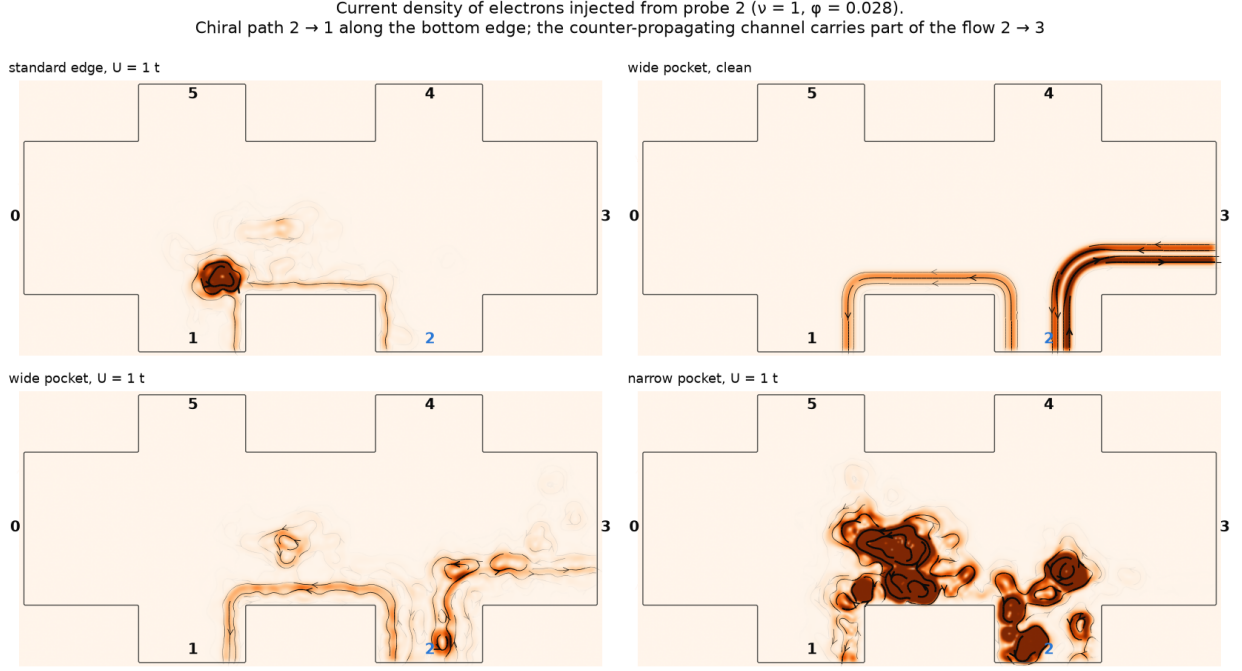


Figure 8. Current density of electrons injected from probe 2 ($\nu = 1$, $\varphi = 0.028$). The colour scale is shared and clipped from above.

- **The $\nu = 1$ window is the main difference between the two measurements.** While the pair overlaps it is localized, and both measurements are quantized. Once the partners are $3-4 l_B$ apart, the Hall bar is already clearly non-quantized ($R_{xx}^{\text{bot}} = 0.09-0.19$, with the reference quantized to 10^{-4}), while G_{2t} is still $0.986-1.000$. In individual realizations R_{xx}^{bot} reaches 0.4 and t_b of individual segments reaches 0.55 .
- From $E_F \approx 0.436$ the reference starts to deviate from quantization, and at $E_F \geq 0.452$ its bulk transition begins, so the comparison is no longer meaningful.

7 Interpretation and comparison with [1, 2]

1. **Same physics, different observables.** By (3) the two-terminal conductance is an integer both for $t_b = 0$ (as if there were no pair) and for $t_b = 1$ (the ballistic pair counts as one more channel); only an intermediate t_b is visible. In the Hall bar, t_b of each segment between neighbouring contacts produces R_{xx} and a difference of Hall voltages, and $t_b = 1$ gives the *largest* deviation. The “breakdown of quantization” of [1] (intermediate t_b near the pocket bottom) and the “protection of quantization by backscattering” of Sec. 5 are two ends of the same dependence of t_b on the pair separation. Which one is observed depends on the edge length between contacts, the disorder strength and the way the measurement is done.
2. **The plateau labels differ.** In [1] “quantized” means $G = \nu + 1$ above the pocket bottom (the ballistic pair counted as a channel), and breakdown means $G < \nu + 1$ within the window Γ_ℓ . For a Hall bar with bulk filling ν , quantization means $R_{xy} = h/\nu e^2$ and $R_{xx} = 0$, i.e. $t_b = 0$.
3. **The four-terminal measurement is more sensitive.** A segment between probes ($60-70 a$) is shorter than the whole edge ($300 a$), and $t_b \sim e^{-L/\xi}$ is larger on it. At $E_F = 0.416-0.432$ G_{2t} differs from 1 by at most 0.015 , while R_{xx}^{bot} of the same sample is $0.09-0.19 h/e^2$.

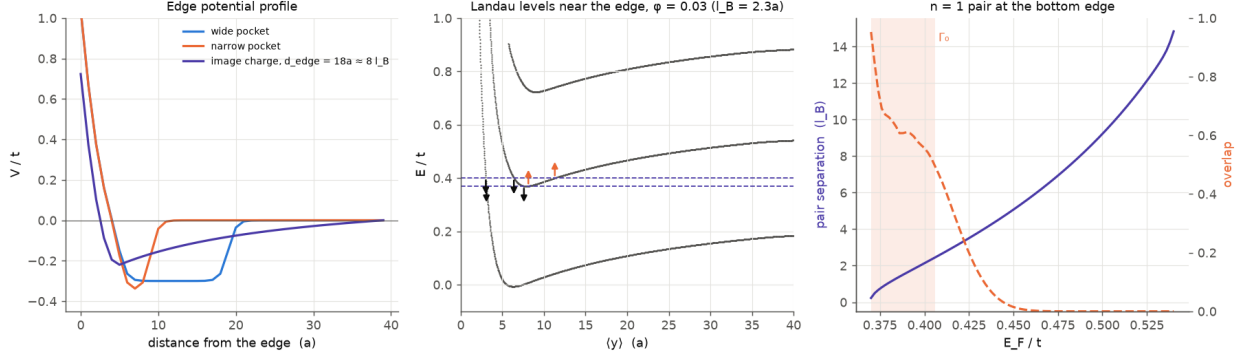


Figure 9. Left: edge profiles (wide and narrow pockets, image-charge potential). Centre: bent Landau levels near the edge with the image-charge potential at $\varphi = 0.03$, and the modes at two Fermi energies. Right: separation and overlap of the $n = 1$ pair as a function of E_F ; the shading marks the window Γ_0 above the pocket bottom.

4. **The comment [2].** We do not test its first objection (no pocket because of the donor background, Fermi-level pinning at the etched surface and a surface-state capacitance $\gg \varepsilon_0/d$): all our conclusions are conditional, *if* a pocket exists. The second objection (the paper does not compute R_{xx} and R_{xy}) is answered directly by our calculation: in a six-terminal Hall bar such a pocket gives $R_{xx} \neq 0$ *only along the edge next to the metal*, different R_{xy} on the two probe pairs (their difference equals R_{xx} of that edge), and deviations only on the low-field side of the plateaus. This distinguishes the mechanism from a bulk one (for instance vacuum-induced), for which R_{xx} appears along both edges.
5. **What of [1] is missing here:** spin and the Zeeman plateaus, finite temperature, fixed particle number. The leads in the code of [1] have neither field nor pocket, ours have both; in both cases the counter-propagating channel ends in a contact, so this does not affect the conclusion.

8 Conclusions and experimental signatures

1. A counter-propagating partner of the chiral channel appears when the edge hosts a pocket about as deep as the distance from E_F to the next Landau level. This happens on the low-field side of every plateau.
2. The intuition that “backscattering on one edge should not spoil the Hall response” is correct, and in a stronger form: backscattering inside an edge cannot break quantization, it protects it. Everything that happens inside an edge between two contacts reduces to t_b , and quantization is broken only for $t_b \neq 0$.
3. Deviations of R_{xy} from $h/\nu e^2$ and $R_{xx} \neq 0$ appear when the counter-propagating channel is *not* backscattered (the partners do not overlap) *and* reaches ohmic contacts. The bulk σ_{xy} stays $\nu e^2/h$: this is an effect of the contacts and the geometry, not a breakdown of topology.
4. With the image-charge potential of [1] the picture is the same. As the partners separate, R_{xx} and the difference of Hall voltages appear in the Hall bar before the two-terminal G deviates.

Experimental signatures of the scenario:

- $R_{xx} \neq 0$ only along the edge that has the pocket (the metal);
- the Hall voltages on different probe pairs differ by exactly R_{xx} of that edge;

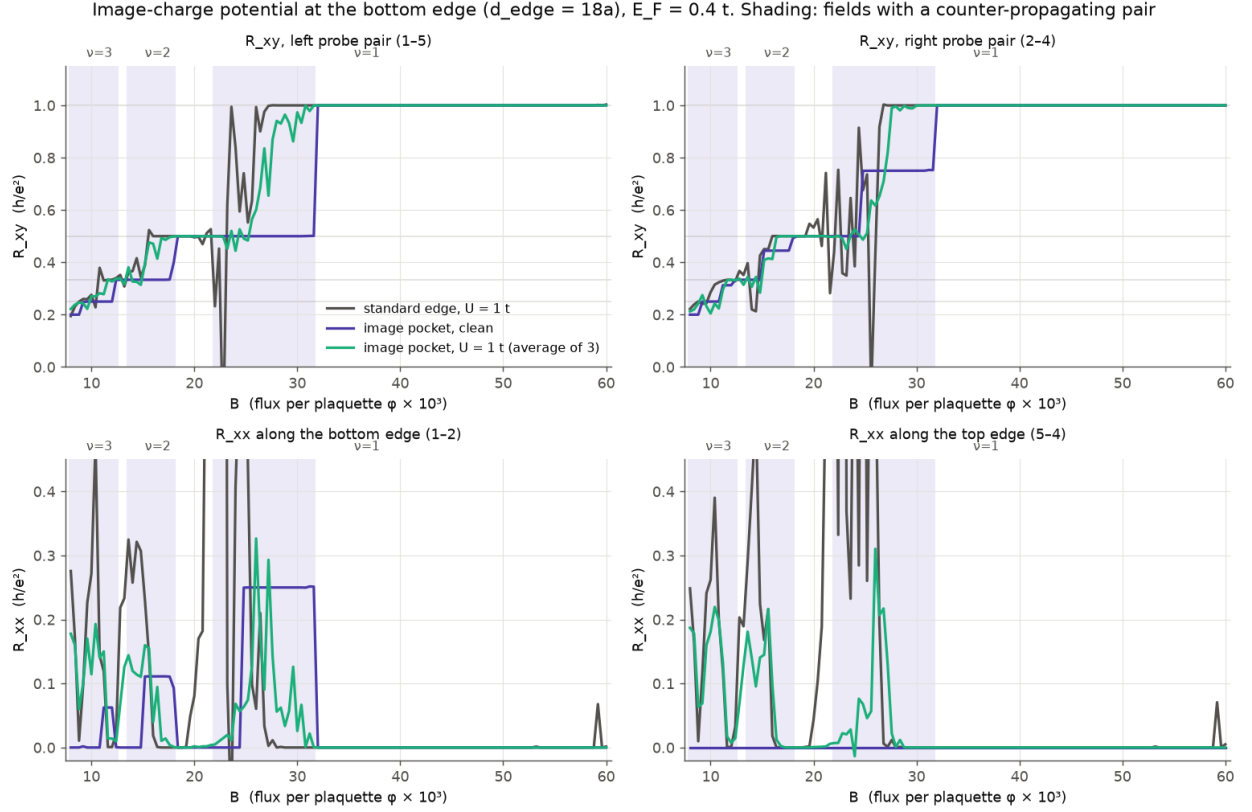


Figure 10. Image-charge potential at the bottom edge, field sweep at $E_F = 0.4t$. Shading: fields with a counter-propagating pair.

- the effect is concentrated on the low-field side of the plateaus and disappears if the pocket does not reach the contacts;
- a two-terminal measurement may barely show it.

9 Limitations

- The model is single-particle and coherent: there is no electron–electron interaction, no edge reconstruction and no inelastic equilibration between channels. Equilibration over lengths longer than the equilibration length also gives $t_b \rightarrow 0$ (as for $\nu = 2/3$).
- The pocket is put in by hand; whether it forms in the self-consistent electrostatics of a real mesa [2, 6] is not addressed here.
- The sample is mesoscopic ($L \approx 125 l_B$), so the transitions between plateaus fluctuate strongly. The main curves are averaged over three disorder realizations, some curves are a single realization.
- The pocket parameters are chosen for clarity, not fitted to a particular experiment.

A Reproducibility

Code and data are in the `qhe_pockets/` directory of the repository: `model.py` (geometry, potentials, transport), `lb.py` (Landauer–Büttiker model), `edge_modes.py` (ribbon modes), `run_sweeps.py`, `width_sweep.py`, `image_sweeps.py` (calculations), `plots.py`, `plots_image.py` (figures), `plots_en.py` (English figures).

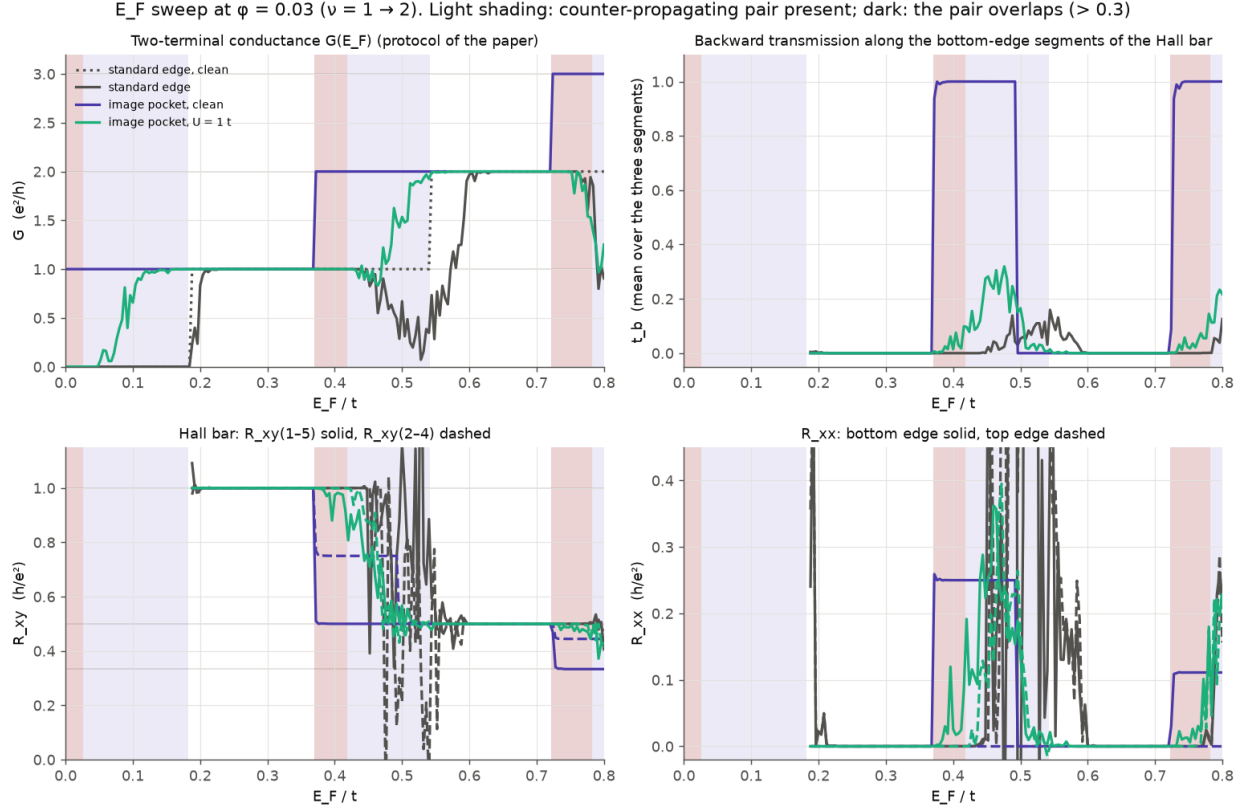


Figure 11. E_F sweep at $\varphi = 0.03$. Light shading: counter-propagating pair present; dark: the pair overlaps ($O > 0.3$). Top left: two-terminal conductance of the plain bar (protocol of [1]); top right: t_b averaged over the three bottom-edge segments of the Hall bar; bottom: Hall-bar resistances.

```
conda install -c conda-forge kwant matplotlib scipy
OMP_NUM_THREADS=1 python run_sweeps.py      # ~1 h on 4 cores
OMP_NUM_THREADS=1 python width_sweep.py     # ~20 min
OMP_NUM_THREADS=1 python image_sweeps.py    # ~20 min
python plots.py && python plots_image.py && python plots_en.py
```

A note on the field calibration: `kwant.physics.magnetic_gauge` needs $B = 2\varphi$ to produce φ flux quanta per plaquette (checked for Kwant 1.5).

References

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